Semiotics and Semantics (Discussion) ~ Lorena Sganzerla ~ Active Inferer

Courses/ActiveInferenceForTheSocialSciences | Lorena Sganzerla ~ Active Inference for Social Sciences 2023 | 1:45:19

Hello and welcome. It is September 12, 2023. We're here in the Active Inference for the
Social Sciences course at the Active Inference Institute and having a discussion section on
semiotics and semantics here with Lorena Siganzala. We're going to be having a fun discussion,
I hope. If you're watching live, please feel free to write any questions in the live chat. If you
thought about joining and you weren't sure, it's not too late. Otherwise, we're going to pick up on
some key points from the lecture and on the topic and see where we go. So, Lorena, welcome back and
let's, however you want to begin, let's go from there. Okay. Hello. Thank you, Daniel.
So, let's try to recap a little bit where we left off during the lecture.
Um, just to remind us where we were, like my point, what I was trying to articulate, uh, was the idea of
what does it mean to talk about a form of semantics in a way that is not really
loaded because it's an interesting thought, like, to have a semantic, semantics that is actually only
formal, right? In principle, it would be relative to meaningful, uh, to meaning, right? And in the notion that
active inference is trying to cache it out.
It is, it's more like an idea or a way of modeling how, how we deal with meaningful situations.
And to understand that in a more interesting, more embodied and more concrete way, I was suggesting
that we could use biosemiotics as, uh, as a model and as a theory for living creatures, like language users to
navigate the realm of symbols.
And gave one example, which came from, uh, James Scott, which is the European forestry that comes with a
practice understanding this scientific understanding of forests as a practice
that may, that allows us to, to refer to meaningful, to meaningful things in the environment in terms of
what is valuable, what's interesting and what is not. And what, interestingly, what, what is more
important also from what becomes valuable, it, it is what is cropped out from the meaningful perspective. So
things, natural, uh, parts of nature becomes left out or understood as something that is not desirable.
And you just understand what you don't want in your articulating principle as weeds or something that you
should get rid of in respect to what is more desirable in terms of crops and how you understand nature in a more
commoditized way.
And I find like this example from Scott kind of very, very, uh, elucidating.
Uh, in, in the book, he creates this, in the book, he's talking about states, how we articulate life around
one specific notion of state. But in this specific example, he has the idea of one official observer,
and this official observer establishes a principle of meaning.
In a way, you have to have a handle of nature.
And that also articulates how villages and how practices around agriculture and, and how people relate to each
other within these practices gets organized.

Specifically for one interesting point that he makes that is to appreciate the constriction of that is
like a vision that notices not what within the meaningful crops, but which is left outside the field of vision.
and i bring that specifically to understand how
the free unity principle can or could be connected to a generalized synchronization
over time in relation to those symbols and those practices in terms of meat construction and
how those how this meat construction brings along with the symbolic use of
language a predictive value
thank you brin wanna bring up any thoughts on that or any other
pieces or i'm happy to add a comment
bryn
okay guess not okay um
a lot to say there there was a little bit of a double play with what is cropped on one hand what is
cropped is the industrial crop that is being prepared and um and then the other side of that coin is what
is cropped out and what what we crop in is the part that we're paying attention to and we can't pay
attention to everything so it's it's like two sides of the same coin what we pay attention to
what we include in attention in our cognitive awareness leaves the shadow which is what we exclude
um um that could be like the weeds or the other ecosystem services so by highlighting the meaning
or one dimension of meaning or one stance on meaning that only it's like a seesaw that becomes salient to
the extent that something else must be deprioritized so that's how we accentuate these distributions of meaning
and what do we get from active inference and and free energy principle beyond for example what's
provided in seeing like a state i think there's a lot we could say on that but one piece is we get a
process theory a real procedural approach to how different systems come to do sense making and cognition
and action so just describing the lay of the land or the history of forestry in europe or um any other
state of affairs in the world merely describing it may not even help us determine what's going to happen
a short time later let alone a long time later or what might happen in a very different ecology
whereas if we understand the processes of sense making and decision making that different kinds of
cognitive agents do certainly humans also though states as cognitive entities which is what seeing
like a state or thinking like a state entails we get a process-based view on what otherwise might be merely
descriptive and along with that process-based view we gain an increased ability to make
unique explanations predictions accounts and so on
yeah we can we we can also understand we can also uh understand it in a different perspective in terms of
realizing what becomes sensitized and what becomes desensitized along the process so you can you can
possibly model that using this kind of tools and understand how that happens over time right i don't
know how how much in terms of prediction of things that were not foreseen in these descriptions we can
we can achieve in fact you can achieve some some level for sure but like that system should be decide how
how how that delivers a very all-encompassing prediction in this kind in these cases they don't believe it
can right i don't think that's the purpose as well but we can at least have which is might not be as

positive as we want but i think we can understand how we can gain some understanding in this process of being more sensitive or less sensitive to specific

primes and how this organization organizational principles uh how powerful they become over time

and how this organization is and how this organization is and how this organization is

yeah i'll read some comments in the chat and then lorena or brin feel free to just give any thoughts

upcycle club wrote hello biomimicry can enhance biosemiotics by showing us how nature solves problems

and creates beauty through signs and codes and by encouraging us to apply these principles to our own

creations yeah the idea of coding is an interesting idea but to talk about codes i don't think we can

that's my take right i don't understand like you have we haven't coded uh any type of principles within our

uh cognitive ability but uh biologically speaking however you can understand these encodings in terms of

becoming more or less sensitive over time to environmental cues as the example in terms of uh i use the example

of forestry right so in that specific case this what becomes more what you become more sensitive and the

more you organize your life or social life is organized around specific principles then you can make sense of

these relational stories uh instead of just leveraging one full uh

difficult to explain notion of encoding in terms of social cognition because it will be hard to explain

how the brain will be encoding these specific symbols but they can be cashed out in the environment

how the brain will be able to determine the environment relationally in terms of those organizational principles

and then you have this idea of biomimicry like this great iterations and how that comes along

and how that comes along with uh to explain what this mimic or this meme will be then well it's not so so straightforward i think

that's what's going to be doing well certainly decompiling the source code of nature

is somewhere between challenging and misguided or impossible but that notion of a code script

for biology has animated discussions of life going back to for example schrodinger's what is life he basically says there's two aspects of living systems there's their organizational capacity their negantropic capacity and then there's their code script their a periodic crystalline informational repository

repository and then another place that coding comes up a little bit less in the source code space is predictive coding and that's in a way highly compatible with active inference and gives a deflationary account on how the semiotics are encoded

rather than saying well this is what this variable in the source code means or this is what this codes for

or here's a code switch predictive coding is like saying we're not even going to touch the semantics of what the sign is itself

we're just going to hypothesize that the way that information about the sign is transferred

is through this maximum information signaling approach called predictive coding because predictive coding arose out of video compression algorithms and only at the end of the 90s

was found to be increasingly relevant for neural circuitry and so on so it's another example of where like a process-based

approach

ends up deflating

or reframing
the
otherwise
semantically loaded question
it's like switching out the question of like where we're going to be with how we're going to get there
and instead of only focusing on the meaning of the signs first off as if the meaning of the signs could be
discussed outside of their
implementation
and then even if we could figure out what the signs were
we would then just turn around and want to know how they were transmitted
and so predictive coding or predictive processing
in a way short circuits that or inverse it by first looking to explain
how information is transmitted
that could support
and then in the context of biological systems we naturally have biosemiotics
yeah
i i would ask the question when you talk about information being transmitted
right what that actually means and i think that's a very very big
topic and a very large discussion in terms of
information
information theory when specifically applying that into
um
biological systems
i think there are some interesting people working on information that
makes the case that shannon has a point
and he's really interested
not in the semantics in the semantics or in the meaning or anything else it's kind of interested in
the problem of transmission and when you talk about transmission
it's worth to ask what is being transmitted exactly in the terms of
of signals even if you're capturing specific patterns and you think that in the department
is this thing that will be transmitted in that sense
of the information
uh
worth asking question
to understand
like the
this notion of what is actually being
transmitted in that sense like

information pro disinformation theory is focusing on this notion of
the problem of transmission
and then all the the concepts that come along to explain that in terms of
uncertainty or noise and everything that
uh impacts this
transmissional processes
and
if you're saying like we're transmitting information in this sense that is breaking this
semantic code apart yes it is because like the problem is not focused on that
it's that focusing on the specific correlations that we have that we call it
with you're mentioning as information and how these correlations
are a ab an issue from from the get-go
from what is being picked up in one side being
correlated to what comes into the other side right how that
how do you lose or gain more information in that
what is being picked up in one side of the other side of the other side of the
the deeper question here is
uh how how can you say that you are actually gaining or losing that information because you
are defining this correlation that could be transmitted or not
from the get-go if you don't have that
you don't have anything being actually actually communicated
so with that in mind you can
explain these concepts in terms of function and then there's a bunch of
theories that navigate the sense of
uh
signaling that is specifically only functional and then predictive coding tries to
try to try to push that a little further maybe
but then you have to explain in in a deep sense how this information is being actually transmitted and
processed and i don't think that is really really as obvious and as clear as we think it is
specifically when we're talking about this biological way of nature navigating this process this signaling
we have to talk about see because the correlation if it's defined i've been issue like from the beginning
you then only have this kind of transmission for someone specific that can be able to understand that
the other side right
well certainly whether we're talking about
a chemical pheromone
that requires some type of receptor or something more ephemeral like language that requires
more of a semantic reference frame
there's no

receiving the message without the sender it's like they're kind of the two sides of the phone call

um i'll ask a question from the chat love evolve wrote how about cultural encoding and principles

in contrast to organizational so how do we think about

cultural meaning and is and is that similar different than organizational

cultural meaning

yeah that's that's a good point right i think if you take very seriously the example from the uh the forest

tree is a practice then then i think it has to be organizational in that sense the meaning

is is that this is

and cannot ignore like the history of this practice is being pushed further and replicated

through generations right so

the way like the villages will be

organized in terms of that specific practice for instance can be understood in a meaningful

and cultural way that organizes life

in virtue of

some specific principles that sensitizes our agents towards specific directions

and I think

I don't know if you get my point

for instance you can talk about

if you think about archaeological findings and how archaeological descriptions of specific

sites

they usually know exactly if you have some kind of history already from this communities how this how that city would be likely to be organized

first off like where is the water and if the water is in certain directions probably there will be certain kinds of relationships with housing and communal housing and where they're going to place their dead in relation to this kind of specific organizational principle the water is here so plantations will be not so far away from that.

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smaller things, inside of bigger things, we can always build something and then we can go, oh wow, it's like there's a lot of top-down influence. Maybe there's a higher order factor or slower factor or a contextual factor we should consider. Or, oh, we were abstracting across these smaller things inside, but for a later day, or if relevant, we can dive in. And so then it allows a light hold or a grasp on the proposed generative model. Obviously, it's map, not territory, but even beyond that, we always know that we can expand laterally and across scales starting from anywhere in an iterated modeling process without our first or even 11th model being proposed as like the accounts of some complex cultural phenomena, which we're always going to only ever have our own perspectives on. Yeah, I think you don't really find that thinking being so all and copious, saying like you have one model that will model them all. You can navigate those models in these more instrumental terms. It raises different questions when you put things like that in terms of genitalology of models and models and theories and sciences, but they can be also described as tools for grasping in describing phenomena in a more non-committal way.
How much that smuggles in philosophical or more ontological problems, it is still, I think, under debate, but it is still

a tool to bring insights to think about specific cultural collective phenomena, which is not, it's complicated to, because I think what I'm getting at, I'm kind of like going back and forth, but when you talk about social sciences and modeling, usually you have like ways of modeling agents or ways of modeling

collectives and active inferences trying to come up with a modeling in which you can understand agency and collectives within the same modeling schema.

And I think the specific, specifically for these cases, if you're interested in understanding types of cultural phenomena among language users and

in specific contexts, you can at least partially understand or gain some insight in this agent collective relationship that

that is a little bit more interesting than we have found so far in the social sciences, because it can have like, or specific understandings of means, but then how the agent, where is the agency in that? You lose a little bit of agency.

When you're trying to understand only agents, then you don't know how that kind of scales up. Because active inference tries and offers this scale free modeling tools that facilitate this, this conversion into agents, agents and collectives.

I would be interested in asking you actually how? How do you navigate that? When you talk about social phenomena in ants or termites or?

permites or because then yeah you have very specific coding demarcations I guess for for this

sort of environments yeah it's a great question it's also whether it's a open wounds or a dirty

closet that's just shut there's actually a lot of ambiguity and unresolved tension around

well is the ant colony social are they social insects that reifies the nest mate as the

individual and then what they're doing is like when we hang out and talk so reifying the nest mate as

individual makes the colony social but if the colony is individual then what nest mates are doing with

each other is not social any more than we would describe the liver and the kidney as having like

a social single cellular interaction but we know that would be missing the point

and so there definitely is a tension with some of these evolutionary lineages like ants being

monophyletically obligate eusocial all and only ants are eusocial they all live in the colonies there's no

free living nest mate whereas other parts of the evolutionary tree there are life histories that do look a little bit more like

social collaboration that can then dissolve and so knowing what level of lock-in

makes something social when are we socializing and what level of lock-in

goes beyond social and it kind of like makes when the blanket is so strong around the social

that like for multicellularity that it it's kind of losing the social feature again because if we just

use social to mean the connective tissue or the kind of lubricant or the spaces between everything

then it means nothing so a real tension is what space does the social account for

and is this going to be a concept with a lot of creep looking well it's a social relationship between

the roots of the tree and the fungus and it's a social relationship amongst the molecules in the

test tube is that too broad of a social conception are there downsides or are there trade-offs associated

with a broader social concept do we dilute what the social means and then on the other side might we exclude

phenomena that could importantly be considered social well that's not social they're just talking on a train
it's like well that's the other side if we were to have carving out encounters that we do want to include
a social and is there a true social does active inference help us sidestep or at least negotiate this territory
because we could develop a generative model or an ecosystem of shared intelligence with one person and
two internet of things devices and a bird and a metronome and just say we made what we made
and then leave it as a second order interpretation whether that is a social setting
i mean i think you you were pointing at like how i think granularity of those scales right if you have a model
that once you want to everything that you already have it's probably not interesting and not informative
because you cannot like define those limits of what you're trying to understand and then on the other
hand when you do that when you define those limits if you're too narrow or you're gonna live out you're
gonna idealize also in a way right and leave out many other parts of of the system that you're interested in
you're interested in but if you don't do that if you don't if you don't make this this if you don't
make this decision when when modeling anything then how how you can have any access to
to the phenomenon right and and to connect this to the forestry so the forestry example
the um deserata was lumber now that may have been a very um narrowly scoped desire but it's it's a
relevant thing to to want this resource and then industrialization science as a process um professionalization
those allowed the description and and the handling and ultimately the kind of top-down um legibility
and control thinking like a forestry company now with respect to the social sciences by analogy to
forestry science so now we're entering an era of increased legibility and control of social systems
and human natural capital and attention as a resource trust as a resource in a way is being cropped
analogous to produce the contents of lumber but now there are social contents so as the social
goes from being just only simply the water we swim in
not something that's even compared counterfactually to alternative cultural possibilities per se but
falls under the scope of the professional management like forestry
then that's when this discussion of well what's the lumber and what is the weeds that are being pruned
and what ecosystem services when we pay attention to certain parts of human social culture and when we seek to
make more of this kind of lumber
what else does that come along with and what decisions
what decisions come into play and that comes along when you see the process
you
or instagram i don't know twitter after what happened to twitter but um
but yeah the in terms of attention uh pruning you do have that very effectively already happening
you already you can see how that happens if efficiently in the world in elections for instance it can be very very easily manipulated and we saw what happened
in elections for instance in brazil or in the us and well all over
uh you navigate a system of uh attention organization that selects in only what it thinks that you're interested in

in the effects of that in terms of decision making and in terms of trust in terms of
uh diversity of discourse we have seen that it can be devastating
so the and this the question would be like how can you make this a little bit how can you use active
imprints and these tools that we have to make this kind of technology a little bit more inclusive
without being super disruptive is it even like
a possibility i i really don't know i think it's an ethical problem that we have when we talk about
social systems and these principles of attention exclusion leaving things out
um a lot to say on that um with modeling and interpolating and models whether it's large language
model type or linear regression type it is able to interpolate to put data points within um an enclosure
of what has already been described and bayesian statistics and other types of prediction algorithms are
not immune to that either if we have some prior distribution belief um we believe it's possible for it to be
between 20 and 30 degrees outside then when we get data inside of our bounds we can converge and that
actually tightens our beliefs not always a bad thing of course and then outside of the scope of the
possible those data are just seen as noise and they may not even be integrated so then that can lead to
a reinforcing process not necessarily an adaptive one either where information that's confirmatory is used
to ratchet and increase precision and then information that is not confirmatory is discarded as not useful
it doesn't yeah in this case is yeah it doesn't even enter the uh the landscape of attention of the user in
is um um

in relation to this idea of forestry is because we have now organizational principles that are
faster and more populated with information and attention-grabbing kinds of prompts.
We transform the same principle into
a harder to grasp process. It becomes more complex, it becomes harder to describe.
This is the process of understanding that in terms of forestry is kind of way more grounded
when you apply that socially to this kind of environment
that you were talking like internet or just highly volatile
and virtual in principle in environments. It becomes much harder to assess what is being
pruned out or not. You just don't know. It makes it more clear that
there is a process of pruning. You have no control of what is being taken out.
You have no idea what's being like left away from your perspective of attention.
So to this topic of intended or unintended semantic pruning, I think this has helped me
identify attention. If we want to talk about the semantics or semiotics, biosemionics, whatever we
want to call it, of the letter e in relationship to letter a, we're looking for an account that might
convey letterness, but we're generalized above enus or anus. So in some ways we've erased or abstractly
generalized away from the particulars of the sign's meaning by taking a semiotic perspective.
And so it's like we got into the game to have a semantic account of biological phenomena or of
any phenomena for biological entity, yet this process in a sense distances us from the particulars that
actually do grant it that meaning in the actuality. Like if we walk into a sacred space and it's like,
oh, well, that is just doing that over there. And this architecture is doing this.
And because of the layout, people must use the room this way.
That's removed from the actual
meaning, which is the cause and consequence with the organization and culture that we discussed.
But that prima facie accounts
doesn't generalize. Yet any generalization that we make so that we can talk about multiple places

or multiple letters on a common accounting framework is going to remove those particularities.

So it's a very challenging catch 22, whether we want to really account for the n equals one

and maybe even live and act and play and work in the n equals one versus develop something that might be satisfying and cut and dry, but it's always going to be on the sidelines with respect to what it actually means.

But what do you mean like is I don't understand what you are getting at like, is it you mean like

generalized or not generalized? I didn't get that.

Like when we have a general, when we generalize, when we ask what words mean, what words mean beyond any given word, we've lost touch with what any given word means.

But words are what they actually mean in a situation.

So it's just in this scientific approach to meaning, semiotics and semantics,

we have these two tensions or a tension to reconcile between actually giving an account of the meaningfulness

the specific thing versus taking a more scientific approach, which would be gathering a group of similar things

and finding their shared attributes,

which out of necessity must abstract, must crop out features that

are empirically or anecdotally taken to not matter.

Like we abstract whether it was a Monday or Tuesday that we did the experiments,

we just combine all of our replicates together when we do the scientific analysis.

You mean,

you mean if you generalize the practice, you lose

what you mean if you generalize it too much, you lose everything that's meaningful to you in terms of like,

you formalize it.

And then you have no, yeah, you have an empty semantics.

You have a semantics with no meaning that can only be meaningful in terms of how you apply that in terms of specific practices.

Yeah. It's like, I have a framework for, you know, the formal meaning of

letters within words, within sentences, within paragraphs. And then it's like, well, but

now what's your next N equals one word that you're actually going to bring?

It's not just about building some abstract scaffold

for bigger and bigger boulders of meaning

at some point through our embodied thought and action.

And then it's like, I think, in our experience, it is going to come down into the particulars.

And it doesn't immediately seem clear that simply generalizing further

brings us there.

Yeah.

Yeah. I don't really believe you can have that kind of generalization. I think we

attempt to build whatever, what I see that's the interesting aspect of the formal semantics.

You try to build this more general

understanding that's just formalizes to model, but for that to be meaningful, it has to be embedded in a way of practicing it. Like in a forestry practice, you have to be embedded in a cultural environment in that can be actually observed in terms of what happens when a generative model is being updated or not.
Is that what you mean? Because otherwise I don't really understand how it becomes too formal, it becomes too abstract in terms of some, I would not understand what that is.
I'll think of a little more. Brynn or TJ, do you want to add any thoughts or questions?
Yeah. Sorry. I, I think like, I don't have a lot of context. I've been like following some of the discussion, but not attended a lot of these calls. I'm like, it's all. Yeah. Yeah.
Um, yeah, I don't know. Uh, I think, um, yeah, I'm in general, just sort of interested in trying to understand the relationship between, um, yeah, like sorts of like constructionist epistemologies and, um, yeah, like generative models and active inference, um, style stuff and, um, how active inference style, like, yeah, what are the philosophical commitments that active inference involves with respect to knowledge and epistemology? And there's a sense in which like rational choice models often assume a certain sort of like, um, objectivity to beliefs and then desires are where most of the degrees of freedom lie or something. Um, whereas there's a lot of like issue, like there's a lot of things that come up once you start talking about reflexivities where concepts themselves become self-fulfilling prophecies, uh, what you believe about what your beliefs about society ends up shaping the society itself. So there's a sense in which there's a score dependence between the object of knowledge and knowledge about that object itself. And I think, yeah, like I'm sort of interested in like how the active entrance ways of thinking about this offers what perspectives on those questions and stuff like that. Um, yeah.
Thank you. Great questions.
Yeah.

active inference.
I think you can think of in terms of this.
Well, there's a few ways to think about that, I guess.
The knowledge itself, it has to come epistemically from where your agent is actually embedded and situated, specifically in active inference.
It's embodied.
It kind of needs to have a model.
The model is modeling a body.
It's modeling a body that is in the world and it's all the time situated.
It's never disconnected to its environment and the history of the world.
That environment, right?
Yeah.
But if you think, if you push that to the object of that knowledge, I don't know how different that will, how can you differentiate that so sharply when you talk about active inference?
Because if you think of in terms of a generative model, what will be updated is updated from a prior that is always directed in this body, in this situated realm.
So they sort of conflated as I see, maybe, maybe I'm mistaken, but they sort of conflate in how you're going to update your generative model, which it will be at the same time.
At the same time, what do you know?
And what are you the object of this knowledge?
So, sorry, if I can just like, raise a question to that.
Yeah.
So I think like, I think that there is a sense in which like, with respect to something that an epistemology would be called self knowledge, there is, there is a question here that the, the active inference agent has this model of that is like situated in the body or like in, in its like, like structure and is also modeling that structure itself.
And there is like this issue of like, where, where, yeah, like the belief is not unidirectionally dependent on the object sort of a thing, but has this feedback loop thing comes up there.
But there's another way in which I think like, where this sort of thing comes where, even if the object is outside of the body of the knowledge, I think that that could still come up.
So for instance, if I live in a, if, if an active inference agent lives in a society of other active inference agents where currently all of them defect on prisoners dilemma, but maybe like on further reflection, they would want to like, again, like, I don't know what further reflection would mean in the active inference sense, but I'm like, just saying in the intuitive sense, like that, what would it mean for them to like go from one fixed point to another fixed point where they're all cooperating or, or, or, or something

like that.

And that, that I think is not clear to me, like how that shift can happen in active

inference, where it's not just purely using the empirical information to keep running the, the, the middle of the, of

the, of, of your like free energy equation, but also recognizing the contingencies and then identifying

those contingencies to make something like shifts from one state to another.

Like if, if I model that the society is a cruel space and if everyone models the society is a cruel space, then we will end up creating a society that is a cruel space.

But if only we could recognize that our belief or our model that society is a cruel space is a contingent belief.

And we could all somehow negotiate or communicate and shift to a different belief about what society is and society is a kind place.

then we will end up creating that kind of society or something.

And that sort of like a belief about the thing that society is, it's not entirely within the agent.

Like it's sort of, it, it, it, it, it has a weird inter subjective character.

So it's not entirely outside of the agent either, but it's not entirely in like internal to the agent either or something.

Yeah.

Yeah.

Yeah.

You can think of that.

Like you're talking about active inference action comes first, right?

You have to, you don't have the knowledge like inside your head waiting for you to, to push that out of the action has to come first.

So if you, if you take that as a principle, an organizing principle, it depends on the contingency of your actions.

The possibilities might be limited by your situated condition, but they are limited.

It doesn't mean that you don't have like a range of options from that state space in which you can go, you can navigate.

So you can navigate it for the good or for the bad, but it starts with the, the, the, the action perception loop.

It starts with the action.

I'll add a few other pieces on that.

TJ, very good point.

Like when we see the pragmatic component of action as self-fulfilling epistemic component of action having to do with gaining knowledge and the salience of information.

But the pragmatic component is about bringing our observations into alignment with our preferences.

And then you identified two related settings.

One of them is like pure intra subjectivity.

Like I expect myself or I find myself or I habitually think about X.

Well, that's kind of a trivial self-fulfillment because thinking about something is its own fulfillment.

And then when we're in this second person or social space, we can agree on little performance art or on a game.

And games give evidence to that, that when people are able to perform communication or message passing such that new expectations or preferences are set, whether more top down, like more seeing like a state or more laterally, more like collective behavior, then those regimes can be embodied.

And then you brought up a very important point about the awareness of the contingencies or where awareness is of adjacent possibles or of counterfactuals.

So the body might be, we could model its morphology as doing some computation, but that doesn't mean that the morphology has an awareness or can explore a counterfactual.

Like bodies are not always exploring every counterfactual because skin is in the game.

But again, mentally, it's very cheap and easy to consider counterfactuals and situational awareness of social settings, especially if it can be not understood, but communicated, opens up a space where we're all playing prisoner's dilemma.

Now there's parameterizations of multi-agent prisoner's dilemma where tragically we just all defect all day and never learn.

There's also parameterizations where you just start all cooperating and never defect.

And then the interesting settings are where there might be like phase transitions or where a perturbation moves the system from one attractor into another attractor.

So there's just a lot of richness.

And that's the work of building generative models and that iterated modeling process to unpack and refine.

And active inference isn't going to have a stance on the prisoner's dilemma.

We can use active inference to describe the prisoner's dilemma, but active inference isn't going to come down on literally even a little two-by-two game theory matrix.

So we can then extrapolate that active inference is not going to come down with an opinion on something significantly more rich and real than a two-by-two game theory matrix.

And so that work is what people enact in applying active inference.

So we can make this kind of ail

improvements or something?

Well, you can, we're assuming that we're trying to improve, right?

We don't know that, but you can model them as improving.

But like following Daniel's lead, you were talking about these specific games and contingencies.

They, they, they, they, this, this transitions from one state to another, specifically if

you're talking about this modeling of social actors and talking about several actors, if

your model involves several of them will not be really linear, right?

You don't have like the easy, smooth transition of improvement.

You can capture over time as modeling, how they might tend to go towards a direction and understand that as improvement, but that takes.

Specific iterations that will not be fully linear, I guess.

Specifically if you want to understand error and improvement instead of trial and error, but trial and adding up into this error, like error improvement.

But when you're modeling specifically with dynamical systems, for instance, you, you, you're not going to see a full linear story.

And I don't think you're going to see that.

Uh, if you're modeling using active inference and multiple agents, agents.

I'll, I'll add one piece there.

So how can we go about modeling multi-scale social goals?

So we could think about different citizens in a region and they, the modeler is going to choose what knowledge to realize in the generative model.

Like what you put into the calculator.

It's going to calculate upon.

Now this happens to be a elaborate stochastic method.

But if we only tell people, yeah, there's just your personal income optimized for that.

Your preference is to have that be higher.

Well, then the simulation is going to unfold a certain way.

If we say, well, there's your personal financial income and then there's your city.

And then there's your region.

Then that opens up through, through different mechanisms.

You know, maybe an individual goes flat or neutral on the personal in service of a larger win at a, at a higher level.

Well, now what if the modeler says, it's not just about the financial income.

Now we're also going to have the secondary criterion.

Well, now different agents could attend differently to those two different.

Outcome variables.

And maybe there's one simulation where 50% only care about this and 50% care about that.

And it works perfect or it doesn't work.

It's like a broad surveying, you know, or Lego set or, or grammar or, or active inference ontology for scoping very broad.

Very broadly and very inclusively.

But all the work is in the real simulation developments and sweeping.

And that's, because very little could be said about a game theory game.

If you didn't know what the parameters were in the matrix, you wouldn't know whether this prisoner's dilemma was going to be like an always cooperate or which agents with which strategies would or wouldn't cooperate.

If you don't know what the matrix looks like.

If you don't know what the matrix looks like.

So before the values have kind of been filled in.

It's just like actually describing the adjacencies and the counterfactuals and the contingencies of our scientific model.

But that's still prior to having the model in hand.

And the model, you know, so now we're looking at that tree in the European forest.

And it's like, well, if I had this tool, then I could cut it down this way.

If I had this tool, I could cut it down this way.

Or I could wait.

It's like, that's alternative actions about that tree.

Now in the social science setting, discussion of different scientific methodologies are like counterfactuals about that social tree.

Yeah.

And specifically, if you're navigating these kinds of game theoretical models, if that's what you're interested in, you can even define in terms of prisoner's dilemma, for instance, specific values to see what happens when everybody deflects and how frequent modulate those frequencies in terms of specific values.

In terms of specific, in multiple iterations, how, how, how are you gonna build up the wage or tendency of deflecting again, if your neighbor deflected once and deflected twice or three times or.

And see how that will accommodate.

And see how that will accommodate.

Over time, how many times this agent will change its behavior until it stops changing the behavior.

So, you know, you're not going to be able to change the behavior.

But that you don't really.

But that's that that again, you use it, you might use a Bayesian understanding of Bayesian model for that, but it not necessarily explain.

It does not necessarily explain the action fully.

I mean, social way, you just you're modeling only the interest of that specific actor.

I have a slightly meta question if I can ask both of you.

I'm not sure how these calls usually go.

I'm curious, what are your interests in active inference?

What do you think like what brings you to active inference?

And like, what do you think?

What do you think active inference adds to our theoretical understanding and in what ways or something?

Okay.

Lorraine, go for it.

Well, I think I was trying to touch upon that before and how we can.

Well, if you're if you're navigating, for instance, behavior in terms of game theory, this kind of modeling for talking about like trust games, for instance, you're going to understand how this.

is to add interactions evolve over time, specifically over interactions, but you don't have an account of change because I have very limited options of actions and.

You don't have an account also on scales, and I think active inference tries to edge a little bit and account on how you can scale that and how.

How you can add more parameters into the game and see how they evolve with a more clear.

Understanding some are richer understanding that, for instance, when you talk about specific interactions in game theory.

So when you that's one one aspect that active inference kind of contributes.

Another point is because we have this.

Parameterization that is specifically in terms of belief updating and also in terms of dynamics.

We are always talking about agents that will be situated and embodied and have a very specific.

way of thinking about synchronization with the environment.

Synchronization with the environment all the time, because when you talk about active inference, the action comes always first.

So you have a less you try to have a less reductive model.

specific.

And when you want to model cultural agents and collective action.

I think that's kind of desirable.

That's what you're looking for.

Is it also your area of interest?

Sorry?

Is it also your area of interest?

Mine or is that Lorena's area of interest?

Lorena's.

What do you mean?

Oh, sorry.

Like my question was also like, what brings you to active inference?

And I think you answered a lot of ways in which active inference adds theoretical value, but I wasn't sure like if that was also your subjective interest or motivation.

It's my subjective interest.

Yeah.

I'm curious to see how that pairs.

Yeah.

I'm curious to see how that pairs.

Yeah.

I'm interested in how that helps us to think of this type of phenomenon.

And I'm very interested in how this relationship from agents to collectives and back again work.

And I think it might give us some.

Well, you give us some very insightful way of modeling these relationships.

And navigating is as you want, because this feature of being scale free is kind of helpful to look at things.

You can define your Markov blanket in terms of what you're interested, right?

And see what happens.

Yeah.

I'll just give a few thoughts, even though also surely some of these thoughts are scattered.

So, you asked TJ, what does active inference add to our theoretical accounts of the social?

It does many things in my view, and that's what makes it so valuable epistemically and pragmatically.

It gives us the expressivity within a scale to describe collective behavior and across scales to describe nested systems.

So, that brings continuity to discussions of the social with broader discussions of the biological, ecological, computational, physical, and so on.

So, that's a multi-scale transdisciplinary element.

Active inference provides us with legible first principles, which can either be taken instrumentally as just heuristics, and also in certain situations can be viewed empirically as justifiable first principles.

And then from those legible first principles, which are not English specific, they're already in a space that has a lot more possibility and promise than, for example, a mere enumeration of important adjectives or nouns.

We can develop lists of important adjectives and nouns when we have a broader cognitive modeling framework.

And so, this methodological turn then supports just a tremendous diversity of thought and work.

Instead of asking us to kind of go through the eye of the needle and get on board with this one ranking of features and then try to apply it more broadly, that's like an overfit before the cart situation.

With ACT-INF, we can expand the methodological discussion, connect it to theoretical physics and mathematics in ways that really haven't been seen before, and then that opens up more than it closes down.

It brings up more questions than it resolves in and of itself, but it opens the questions that can be resolved.

Like, what generative model can I make that will keep the temperature in this building livable?

But ACT-INF isn't going to have a stance on what you should do for that building.

So, we need to, in a way, take a step in the methodological direction, which I see as moving towards ACT-INF to enable a million steps in different directions.

I have a bunch of questions here.

I think one is probably just a reference question.

There are three topics that I'll say, if you guys have any papers or articles on those things, I would love to, like, get a reference.

One is, in what ways can we model a collection or an aggregation of active inferences as a, like,

super agent or, like, super, like, active inference super agent or something?

And, like, how does this, like, decomposition and aggregation, like, flexibility work in different contexts?

Like, if there's, like, work that tries to either make informal commentary or formal modeling either ways, I would love to see that.

Secondly, if there are, like, active...

Just quick, on the first...

Just on the first...

Check out the previous section of this course on collective behavior where I reviewed many such works.

Great, thanks.

Okay.

Is it on the same notion?

Is it linked from the same notion thing?

All in the same places and in the same playlist with this course, same syllabus.

Yes.

Got it.

Okay.

So, the second is, is there an active inference account of, say, markets or economic interaction, where, like, your prices act as some sorts of, like, information signals and, like, can we use active inference to then talk about aggregation of market entities or something?

I'm guessing this would also be covered in the same list.

Indeed, people have explored cognitive economics and people are working on it, but that doesn't mean it's a done deal, theoretically or especially in practice.

So, keep going.

It's quite experimental still, but there are a bunch of people working on that, I guess.

So, I don't remember any specific paper right now, but I can look it up and forward to you, but yeah.

Yeah, it would be great.

I can drop my email address here.

Yeah, sure.

The third is probably more trickier than both of these, which is, is there an active inference account of natural language?

Yes, thankfully, colleagues like Elliot Murphy and others have explored specifically the linguistic aspects of natural speech.

I think Lorena's lecture and area of focus generalizes a bit beyond specific natural language because we're talking about semiotics and semantics in a more intermodal sense.

However, yes, there's been limited work from a computational linguistics and neurobiology perspective on speech.

Okay.

So, then...

Okay.

So, you said something like first principles stuff when you're making a commentary on what active inference brings.

I have still, I have, I'm still often like struggling what is active inference offering and I like try to encounter different things.

I think I enjoy a lot of philosophical questions that the conversation opens up, but like, I'm still like really ambivalent about the theory.

Yeah.

And when I read like this new book, which Friston has not written, but like is a co-author, I don't remember what the name of the book.

I don't, yeah, I don't exactly remember.

Right.

Which is like supposedly, like supposed to be the canon and like Friston is a co-author, but in the preface he says.

So that's the 2022 par Friston-Pazullo textbook.

Right.

Right.

Par.

Yeah.

Yeah.

I wouldn't quite say it's the canon since it's a short book, but certainly it is a landmark moment for the fields.

Right.

And when I read that, the one thing that bothered me was it brings up the Helmers, uh, free energy thing, but it's sort of argues that by fiat, like, okay, this is a simple principle.

It can model a lot of things, but that doesn't tell me why I should, why I should trust it.

Like, what, what is the reason to act?

Why, why does this principle lead to cognition?

Like, what, like, sure, you've posited a very simple principle and then said, Hey, look, this simple principle can explain so many things, but well, it can, it also explain things that don't look like cognition.

And like, like, I'm not sure.

Like there's an, I got an account of why that principle is like, uh, works or something.

I'll think of some other ways to say it.

Yeah.

You, you, you can think of like modeling by construction and modeling by first principles and if modeling by first principles, you have, you have some limitations in which you cannot fully explain some things, but the virtue of the model and how the first principle organizes everything that comes along with the model.

Usually might be sufficient or might be a useful, a useful tool to help you explain that phenomena that based on the first principle that you were bringing along.

And that's one way of one part of answering this question.

I think it's might be a big question for the, for the few.

Yeah.

Here, here's some short little threads on that.

Yeah.

We can describe maladaptive cognitive systems or adaptive cognitive systems, whether something's adaptive or not is always in relationship to its niche.

Right.

Like the foraging algorithm that works in the desert doesn't necessarily work in the rainforest.

Right.

So these systems are not like intrinsically adaptive or not.

We want to have natural language that allows sentences that are valid and invalid and have different kinds of semantics.

So again, this is just an expressivity framework that enables people to build cognitive models of high reliability or important settings.

Or you could make something that looks just like a beautiful circuit board and not even worry about what the outcome is.

So, so yeah, for sure.

You can describe all kinds of systems that would be like having a less than functional methodological system.

If we said, well, we're doing a linear regression framework, but we only want positive regressions.

It's like, well, not the methodological level.

We want it to be able to do any slope.

So here in the cognitive modeling setting, we want more expressivity so that we can actually construct the ones that we're interested in.

And then how does this really do cognition?

How is it that a first principles like surprise minimization or the bounding of surprise through free energy, how can that really cover so many cognitive phenomena?

That's the question and the play.

But how is it that Newton's laws of motion can describe so many objects being dropped off a tower?

How is it that Bayesian mechanics can describe so many different priors being updated?

And so by understanding cognitive paths as paths of least action, which is the free energy principle, a wide variety of models become of a similar kind.

Now, whether that's useful to a given person in a given setting, like if you make an active inference model of your business, it doesn't mean you're going to succeed.

If you make a self-driving car with active inference, it doesn't mean it's going to work.

This isn't a guarantee of success, but it's a method that can be utilized.

And we know that the method reaches different kinds of modeling outcomes and supports subsequent perspectives and models from where it reaches.

But we can't a priori limit where the method reaches to like only the successful ones or only biological systems.

Then we wouldn't even have the expressivity to talk about cognitive ecosystems with synthetic intelligences.

And when you're talking about cognition also, we can use this kind of first principles to model aspects of cognition, but you might not necessarily think that cognition is all encompassing like that everywhere, all the same.

But you can use that to model specific things that you're interested in, in cognitive behavior, like adaptivity.

Does it give me any insight to understand what is adaptive behavior, for instance?

And which kind of insights do I get?

Are they interesting?

Are they useful?

So that's what the availability that this kind of model bring along.

So you can, you don't have to believe in, in the whole story, but it gives you a lot of leverage to ask, ask questions and what kind of questions you will be asking.

Hmm.

Okay.

So let's say that like free, like active inference and free energy allows me to talk about systems.

But does it give also give me vocabulary to then classify those systems in terms of, let's say, adaptivity would be one property, but like there could be other properties in terms of like how much, how rich the world model is or stuff like that.

Or, yeah.

Yeah.

If you have a portfolio of generative models, you could summarize them according to the TJ statistic.

You could summarize them according to the number of nodes.

You could put them into a common environment or different environments and summarize their activity in any number of ways.

But we can't talk about a ranking or a summarization of a portfolio of generative models that's not in hand.

So this is a framework that lets us build those diverse generative models.

And then in a very open ended way, do model comparison, model selection, model adequacy, all of the kinds of complex systems, engineering methods that we want.

And also I would ask you, where do you see a peer or a comparison first principle?

Active inference has legible first principles, which are reflected in the active inference ontology and expressions using it, which can be represented in natural language.

So that's the talking about it, but it's not only represented in natural language.

Other frameworks have other first principles that may or may not be legible and people may choose different paths to go down for whichever number of reasons.

Well, you can compare them, right?

Models like they have the virtue, you can compare them and see which one gives you the best insight that you're looking for.

Yeah, I think something that I could compare to active inference is the Bayesian utility maximization thing, which has a different ontology.

different ontology i'm not a fan of that either like that also i'll just note that bayesian utility

maximization is a special case of free energy where there's no epistemic value all you have

is pragmatic value and so a special diminished not necessarily you have value of information

and you have explore exploit trade-offs in bayesian utility maximization as well like you

like like if you load into your utility function if you can beg the question and load epistem

into pragma you're right then all you need is pragma but in expected free energy we have epistemic and pragmatic value which is what enables us this expressivity but i think we're aligned on that
right i i think like i'm somewhat like also slightly confused because in the bayesian utility
like between the languages comparing the languages as such like in the bayesian utility maximization
the exploration parts that are are prioritized based on your x anti uh estimation on the value of
information uh whereas i'm not sure that all holds true for the active entrance model as well that the
the epistemic exploration is prioritized or valued in terms of some of the same uh pragmatic thing as well
you'll you'll you'll have a a lot to explore and learn you can make a model that is 80 20 or 20 80 or 100
zero between epistemic and pragmatic there's nothing we can say across generative models about whether
active inference prioritizes epistem or not because you can make a model that's a hundred percent or zero
sorry i i meant more like does active inference allow us to talk about systems that the bayesian utility
maximization doesn't allow us to talk about like i'm talking particularly about expressive power of the
language yeah um like is it like are the things that the bayesian utility maximization can talk about that
active inference cannot other things that active inference yeah talk about that basically maximization
cannot two short answers one short answer utility maximization must cram epistemic value into pragmatic
value we gain more expressivity when we can articulate out pragmatic value which is the alignment of
preferences with observations from epistemic value which is information gain so that's a huge articulation
and the second thing is when you're in a anything maximization framework when you're in a utility
function now again begging the question how you constructed this function that you're trying to
maximize when you're in a maximization framework we have a well-known set of approaches for finding
local and global maxima using optimization or maximization frameworks what active inference uniquely provides
is it describes the path that that cognitive thing takes as a path of least action that minimizes surprise
so instead of climbing to the top of the hill up are we on a foothill or are we on the biggest hill
we're like a ball rolling to the bottom of the hill finding the least surprising path now that least
surprising path can still include novelty and information gain but it turns out to be a lot more tractable
and connected with statistical physics and quantum mechanics when we talk about the path of least
action ball rolling down the hill rather than this sort of question bag on top of a mountain climbing
adventure bayesian utility approach but it's up to each person to feel how they want to feel yeah
and i think it's a little a little bit what's in the book as well when you take the low and high
road for free energy principle right because how much you want to have your utility maximization function
being uh disconnected to the pragma to do to the action
because when you're modeling behavior it will be hard to do that independently like you have a lot of you
have way more assumptions you're committing to much to a greater level of assumptions for your utility
function if you don't do that
because with the pragma with with the action first that pragmatizes you you you have reasons to describe
have reasons to believe why that information game works the way that it does
you can leverage some explanations on that without it you you have you have you you have a model that

has more assumptions and i think if you have a model with less assumptions it's always healthier let's say so
yeah so i think like that that that does talk about the compare comparative analysis of the expressivity
but i i would also say as a meta point that like i'm not happy with either of those languages so i'm like
i'm suspect that both of those languages claim certain ontological assumptions and i'm like always
suspicious but wait but like where are these ontological assumptions coming from
a few other points we tap into an absolutely meaningful nexus of statistical methods and message
passing schemes with active inference if you can state something in the active inference ontology
and construct a generative model then message passing algorithms can be deployed that contractively
compute that model so that's absolutely non-trivial otherwise it's possible just to
get wrapped up in some analytical formalism but then once you want to do that on a 4k video
now you're on square zero with the computer engineering this is not that way and then i will
respond to your meta point with a meta point which is when we've enumerated the alternatives
the one that we like best personally i feel happy with i'm also open to new alternatives arising but
being unhappy when all things considered is in play why what bothers you it's specifically on the ontology of
uh the active events yeah like i i i think like i'm still looking for i i don't know what exactly i'm
looking for but i'm i feel dissatisfied with the account given for why helmo's free energy uh least action
thing like is the central thing here or something like why does it work like why does this principle yeah
uh
there's a lot of punny answers i could give there yeah that that delta that discrepancy
that's curiosity and that's the space of the open and that's learning and that's your unique contribution to make
and that's everything so that's not just a thorn in your shoe that's like
that's what i think is that's what i think is that it's like our experience of action amidst uncertainty
which is the water we're in and so whether we take a higher order narrative perspective on that
discrepancy being dissatisfying or that discrepancy being satisfying i think it's pretty cool i just need
to minimize my surprise there yeah and you can choose to minimize your surprise through your attention
you can choose to minimize surprise at a higher narrative order like i'm satisfied and complacent
that i don't understand number theory but i use numbers to count objects around me so that discrepancy
and understanding knowing that there's a thousand years of research that could be done on numbers
it doesn't unsettle me but for someone else that might be unsettling and that might lead them to
ruminate it also might lead them to do a phd in number theory so just different paths for different
individuals and contexts but there's always so many technical and meta levels with active huh
okay so i don't want to like keep you guys over time uh but i would ask one last reference based
thing which is um are there active inference accounts of seeing like a state phenomena uh in
any literature i think i heard you mention it once so i was like yeah like of this
yeah check out live stream 33 series but also this course that we're literally actually enacting in
states states

states states

Okay. Where is the 33 thing that you mentioned?

If you go to the page with all of our institute live streams and look in the live stream series, like the paper discussions in the 33 series, there's a .0 video with background and context.

And then we have a .1 and a .2 conversation with the authors.

But Avell, who's also the course coordinator for this course and Kairos research, like Avell's and colleagues' work has heavily built on the seeing like a state, thinking like a state, optimal grasp, all these different topics.

Yeah, I'm on your YouTube channel. I'm slightly finding it hard to find the thing.

If you look up live stream 033.0, or if you go to in the video description of any video where it says all live streams here, go there and search for it.

But you'll find it and you can always email us or get in contact.

But yeah.

And I think Avell's paper, like thinking like a state also can help you a lot to understand how active inference modeling is useful for the social sciences.

And so that's in comparison to other types of modeling.

He makes a comprehensive description.

Interesting.

Great.

Thanks.

And this is our first iteration on this course.

We are going to continue bringing new information to the table and supporting individuals who want to build generative models, you know, around this area.

Whether it's part of an internship program or part of a course credit or different kinds of programs that we'll be able to develop.

But this is the real thing.

And we're able to work with people who want to think and do.

Makes sense.

Indeed.

Any closing thoughts on TJ?

Then I will go.

And then Lorena with the last word.

No, this was great.

Thanks.

Thanks for the conversation.

Thank you.

Well, thank you for Brennan, TJ for joining Lorena for the lecture.

This was a fun discussion.

We never really know where or how it's going to go.

But I'm glad that we could cover so many of these topics and probably raise more questions and footholds than door slams.

So thank you again, Lorena.

Yeah.

Thank you for inviting me.

And I think Active Inference, like, holds a lot of space for experimenting as well.

So some things, you try to see if they make sense and if they're going to hold up and open new questions.

And that might be an interesting aspect of the framework itself.

Thank you, everybody.

Thank you, everybody, for the great live chat comments.

There's a lot there.

I couldn't read it all.

So until next time.

Bye.

Bye.